Solutions to Monthly Online Test - 2 | JEE - 2022

PHYSICS

1.(B) $\vec{\tau} = \begin{vmatrix} i & j & k \\ 3 & 2 & 3 \\ 2 & -3 & 4 \end{vmatrix}$

$\vec{\tau} = (8+9)\hat{i} + (6-12)\hat{j} + (-9-4)\hat{k} = 17\hat{i} - 6\hat{j} - 13\hat{k}$

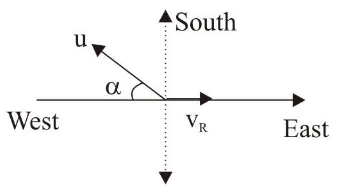
2.(D) $R = \sqrt{A^2 + B^2}$ if $\vec{A} \perp \vec{B} \Rightarrow \theta = 90°$

3.(D) $\vec{B} \times \vec{A} \perp \vec{A} \Rightarrow (\vec{B} \times \vec{A}).\vec{A}$ is zero.

4.(D) Let initial vel $= u$ and $v^2 = u^2 - 2gh \Rightarrow u^2 = 10^2 + 2 \times 10 \times 15 \Rightarrow u = 20\, m/s$

At $t = 3$sec, $v = u - gt \Rightarrow v = 20 - 3 \times 10 = -10\, m/s$ $(i.e. 10\, m/s\ down)$

5.(B)



$u \cos\alpha = v_R \Rightarrow \cos\alpha = 60°, i.e.,$ 30 west of south

6.(C) $0 - t_1 \rightarrow$ uniformly retarded motion

$t_1 - t_2 \rightarrow$ particle at rest

$t_2 - t_3 \rightarrow$ uniform negative velocity

$t_3 - t_4 \rightarrow$ particle at rest

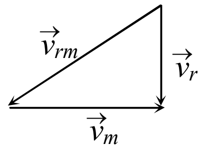
$t_4 - t_5 \rightarrow$ uniform negative velocity

7.(B) $v = \alpha\sqrt{x} \Rightarrow \dfrac{dx}{dt} = \alpha x^{1/2}$, $\quad x^{-1/2}dx = \alpha\, dt$,

$\int_0^x x^{-1/2}dx = \alpha \int_0^t dt$, $\quad 2\sqrt{x} = \alpha t$, $\quad x \propto t^2$

8.(A) $\dfrac{v_A}{v_B} = \dfrac{1/\tan 53°}{1/\tan 30°} = \dfrac{3/4}{\sqrt{3}} = \dfrac{\sqrt{3}}{4}$

9.(D) $v_{rm} = \sqrt{v_r^2 + v_m^2} =$ 5 km/hr



10.(C) $a = \dfrac{dv}{dt}$, $v = \int a\, dt = \int_0^t (kt + c)dt$, $\quad v = \dfrac{kt^2}{2} + ct$

11.(C) $S_r = u_r t + \frac{1}{2} a_r t^2$; $\quad 0 = ut - \frac{1}{2}(g+a)t^2 \quad \Rightarrow \quad a = \frac{2u - gt}{t}$

12.(B) $t = \frac{d}{\sqrt{u_m^2 - u_r^2}} = \frac{\frac{1}{2}}{\sqrt{4^2 - 3^2}} = \frac{1}{2\sqrt{7}}$ hr

13.(C) Height of first body after time t, $h_1 = v_0 t - \frac{1}{2} g t^2$

Height of second body after time $(t - \tau)$, $h_2 = v_0 (t - \tau) - \frac{1}{2} g (t - \tau)^2$

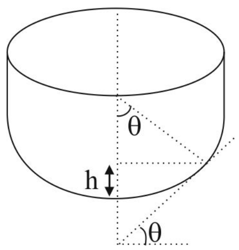
If they meet after time t, $h_1 = h_2 \quad \Rightarrow \quad \tau = \frac{v_0}{g} + \frac{\tau}{2}$

14.(C) $16 = 8t - \frac{1}{2} \times 2t^2$ (equation relative to bus)

$t = 4$s

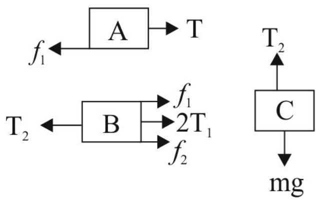
15.(B) $mg\sin\theta + \mu mg\cos\theta = 2\left(mg\sin\theta - \mu mg\cos\theta\right) \Rightarrow \tan\theta = 3\mu$

16.(B)



$\tan\theta = \mu$, $\mu = R\left(1 - \cos\theta\right) = R\left(1 - \frac{\sqrt{3}}{2}\right)$

17.(C)



For equation $T_2 = mg = f_1 + f_2 + 2T_1$

$T_1 = f_1 \Rightarrow mg = 3f_1 + f_2$

$f_1 = \mu m_A g, f_2 = \mu\left(m_B + m_A\right) g$

$f_1 = 0.3 \times 50 \times 10 = 150N, f_2 = 0.3\left(50 + 70\right) \times 10 = 360N$

$\Rightarrow m = 81kg$

18.(A) The equation of velocity is $v = \frac{1}{10}s + 3$

$$a = v\frac{dv}{ds} = \left(\frac{1}{10}s + 3\right)\left(\frac{1}{10}\right),\ a = \frac{1}{100}s + \frac{3}{10}$$

19.(B) Resultant of $\vec{A}\ \&\ \vec{B}$

$$\vec{R} = (4+8)\hat{i} + (-3+8)\hat{j} = 12\hat{i} + 5\hat{j}$$

Unit vector $\hat{R} = \frac{\vec{R}}{R1} = \frac{12\hat{i} + 5\hat{j}}{\sqrt{12^2 + 5^2}}$

$$\hat{R} = \frac{12\hat{i} + 5\hat{j}}{13}$$

20.(C) For $\perp^r$ vector dot product is zero

i.e., $A_x B_x + A_y B_y + A_z B_z = 0$

$$2 \times 4 + 3 \times (-4) + 8 \times \alpha = 0$$

$$\alpha = 1/2$$

Integer Type

21.(12) $v = \frac{ds}{dt}$ for $s = 0$ *to* $s = 60$

$$v = 3 + \frac{(15-3)}{60}s = 3 + \frac{s}{5} \Rightarrow \frac{ds}{v} = dt,\ \int \frac{ds}{3 + \frac{s}{5}} = \int_0^{t_1} dt$$

$$\ln\left|\frac{s+15}{15}\right| = \frac{t_1}{5} \Rightarrow t_1 = 5\ln 5 = 8s$$

For $s = 60$ *to* $s = 120$

$$t_2 = \frac{\Delta s}{v} = \frac{60}{15} = 4s \Rightarrow t = 4 + 8 = 12s$$

22.(75) When second ball is shot 1st ball is at height of $\left(h = 40 \times 2 - \frac{1}{2} \times 10 \times 2^2 = 60m\right)$ and

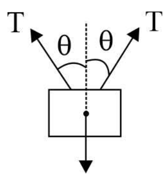
$$v = 40 - 10 \times 2 = 20\,m/s$$

Relative speed $= 40 - 20 = 20\,m/s$

$$\text{time} = \frac{60}{20} = 3\,\text{sec}$$

Height where they collide $= 40 \times 3 - \frac{1}{2} \times 10 \times 3^2 = 75\,m$

23.(1) FBD of $\sqrt{2}m$



$\Rightarrow \quad 2T\cos\theta = \sqrt{2}mg$

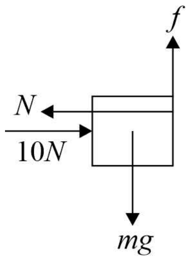
where T = mg

$\sqrt{2}mg$

$\Rightarrow \cos\theta = 1/\sqrt{2}$

$\theta = 45°$

24.(2)



$f = \mu N$

Also, f = mg (if block does not fall)

$\Rightarrow$ mg = 0.2×10 = 2N

25.(2) A + B = 7N, A – B = 3N

$\Rightarrow$ B = 2N

CHEMISTRY

1.(A) $2Ca + O_2 \longrightarrow 2CaO$

$112\,g$ of $CaO \equiv \frac{112}{56}$ mol

i.e. 2 mol so 2 mol of Ca and 1 mol of O_2 are required.

1 mole of O_2 at STP is 22.4 L

2.(D) meq of $AgNO_3 = N \times V\,mL = 0.1 \times 500 = 50\,mmol$ of $AgNO_3$

meq of $KCl = N \times V\,mL = 0.1 \times 500 = 50\,mmol$ of KCl

	$AgNO_3(aq)$	+	$KCl(aq)$	$\longrightarrow$	$AgCl\downarrow$	+	$KNO_3(aq)$
Initial	50 mmol		50 mmol		–		–
Final	–		–		50 mmol		50 mmol

$$[NO_3^-] = \frac{mmol}{Total\ V\,mL} = \frac{50}{1000} = 0.05\,M$$

$$N = M \times x = 0.05 \times 1 = 0.05\,N$$

3.(B) g meq of Na_2CO_3 required $= N \times V_L = 1 \times 5 = 5$

$\frac{g}{E} = g\,meq \quad \Rightarrow \quad g = g\,meq \times E = 5 \times \frac{106}{2} = 265\,g$

Since, the sample is 98% pure, $g \times \frac{98}{100} = 265 \quad \Rightarrow \quad g = 270.41\,g$

4.(B) Since the required conc. lies between the given concentrations, for maximum volume, the entire volume of solution having lower conc. is taken.

So $\quad M_1V_1 + M_2V_2 = M_3V_3$ & $V_1 + V_2 = V_3$

$$M_1V_1 + M_2(V_3 - V_1) = M_3V_3$$

$$(2 \times 1) + 5(V_3 - 1) = 3 \times V_3 \quad \Rightarrow \quad V_3 = \frac{3}{2} = 1.5\,L$$

5.(D) Let p mmol of KOH and q mmol of Na_2CO_3 be present in the original mixture.

For phenolphthalein end point

$$p + q = 20 \times \frac{1}{20}$$

For methyl orange end point

$$p + 2q = 30 \times \frac{1}{20}$$

So $\quad q = 0.5$ and $p = 0.5$

$1000 \times \frac{g}{M_0} = p \quad \Rightarrow \quad \frac{g}{56} \times 1000 = 0.5 \quad \Rightarrow \quad g = \frac{28.0}{1000}g = 0.028\,g$

6.(B) Mass of solute (original) $= \frac{20}{100} \times 400 = 80\,g$

Mass of solute ppt $= 50\,g$

Mass of solute left in solution $= 30\,g$

Mass of solution left $= 400 - 50 = 350\,g$

(w/w) % of solution $= \frac{30}{350} \times 100 = 8.57\%$

7.(B) $Zn + H_2SO_4 \longrightarrow H_2 + ZnSO_4$

$$M = \frac{10\,xd}{M_0} = \frac{10 \times 49 \times 2}{98} = 10$$

mmol of $H_2SO_4 = M \times V_{mL} = 10 \times 50 = 500\,mmol = 500$ mmol of H_2

$22400\,mL \longrightarrow 1000$ mmol

$500\,mmol \longrightarrow \frac{22400}{2}\,mL$ or $11200\,mL = 11.2\,L$

8.(C) $MCl_x \quad + \quad x\,AgNO_3 \quad \longrightarrow \quad x\,AgCl \quad + \quad M(NO_3)_x$

mmol $\quad 100 \times 0.1 = 10 \quad 50 \times 0.6 = 30$

1 mol MCl_x requires x mol $AgNO_3$

10 mmol MCl_x requires 10x mmol $AgNO_3$

$10x = 30 \quad \Rightarrow \quad x = 3$

9.(A) $KClO_3 \xrightarrow{\Delta} KCl + \frac{3}{2}O_2$

$2Al + \frac{3}{2}O_2 \longrightarrow Al_2O_3$

Finally, 1 mole of Al_2O_3 is produced.

10.(A) Magnetic moment $= \sqrt{n(n+2)}\,B.M = 3.87 \quad \Rightarrow \quad n = 3$

11.(C) $\frac{hc}{\lambda} = \frac{hc}{\lambda_0} + \frac{1}{2}mv^2$

$$\Rightarrow \quad v = \sqrt{\frac{2hc}{m}\left(\frac{1}{\lambda} - \frac{1}{\lambda_0}\right)} = \sqrt{\frac{2hc}{m}\left(\frac{\lambda_0 - \lambda}{\lambda\lambda_0}\right)}$$

So, $\quad k = \frac{c}{\lambda\lambda_0}$

12.(D) 3 peaks & 2 radial nodes corresponds to e^- in 3s and Na has valence e^- present in 3s.

13.(D) $\bar{v} = RZ^2\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right) = RZ^2\left(\frac{1}{2^2} - \frac{1}{\infty^2}\right) = RZ^2\left(\frac{1}{4}\right) = \frac{109737 \times 1^2}{4}cm^{-1} = 27434.25\,cm^{-1}$

14.(D) For $Z = 24$,

e^- config is : $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5 4s^1$

For M shell, $n = 3$

Total e^- in $3s + 3p + 3d = 13$

15.(D) For 4p,

$n = 4 \quad , \quad \ell = 1 \quad , \quad m = -1 \text{ or } 0 \text{ or } +1 \quad , \quad s = +\frac{1}{2} \text{ or } -\frac{1}{2}$

16.(D) $\text{I.E.} = +13.6Z^2 \; ; \; \frac{(\text{I.E.})_H}{(\text{I.E.})_{Be^{3+}}} = \frac{Z_H^2}{Z_{Be^{3+}}^2} = \frac{1}{(4)^2} = \frac{1}{16}$

17.(C) 'm' is associated with spatial orientation of the orbital.

18.(B) $\lambda = \frac{h}{p} = \frac{h}{mv} = \frac{6.6 \times 10^{-34} \text{Js}}{0.12 \text{kg} \times 20 \text{m/s}} = 2.75 \times 10^{-34} \text{m}$

19.(C) 3s has 2 nodes

20.(B) $\boxed{\uparrow\downarrow}$ $\boxed{\uparrow\downarrow}\boxed{\uparrow\downarrow}\boxed{\uparrow}$ this arrangement is correct.

2s 2p

<u>Integer Type</u>

21.(5) meq of $H_2SO_4 =$ meq of $NaOH = \frac{1}{2} \times 20 = 10$

mmol of $H_2SO_4 = \frac{\text{meq}}{\text{n - factor}} = \frac{10}{2} = 5$

mmol of $H_2SO_4 \equiv$ mmol of $SO_4^{2-} \equiv$ mmol of $CuSO_4 \cdot x H_2O$

So $\quad \text{mmol} = \frac{g}{M_0} \times 1000$

$5 = \frac{1.2475 \times 1000}{159.5 + 18x} \quad \Rightarrow \quad x = 5$

22.(8) Number of atoms of C in $3g = \frac{g}{M_0} \times N_A = \frac{3}{12} \times N_A$

Number of atoms of S in 'x'g $= \frac{x}{M_0} \times N_A$

$\frac{x}{32} \times N_A = \frac{3}{12} \times N_A \quad \Rightarrow \quad x = 8$

23.(1) 1000 g of salt cake(impure) contains 852 g of Na_2SO_4 and rest are impurities

mol of $Na_2SO_4 = \frac{g}{M_0} = \frac{852}{142}$ mol

mol of NaCl consumed $= 2 \times \frac{852}{142}$ mol

Mass of NaCl consumed $= 2 \times 6 \times 58.5$

$$g \times \frac{70.2}{100} = 2 \times 58.5 \times 6 \quad \Rightarrow \quad g = 1000\,g \text{ or } 1\text{ kg}$$

24.(3) Number of angular nodes $= \ell = 3$ (for f, $\ell = 3$)

25.(56) At STP, $22.4\,L \equiv 1\,mol$

$$11.2\,L \equiv \frac{1}{2}\,mol$$

$2KOH + CO_2 \longrightarrow K_2CO_3 + H_2O$

1 mol ½ mol

Mass of 1 mol KOH = 56 g

MATHEMATICS

1.(B) Clearly $\Rightarrow \alpha^2-\alpha-1=0 \Rightarrow \alpha^2=\alpha+1 \quad \Rightarrow \quad \alpha^3=\alpha^2+\alpha=\alpha+1+\alpha=2\alpha+1$

$\Rightarrow \qquad \alpha^4=2\alpha^2+\alpha=2(\alpha+1)+\alpha=3\alpha+2$

$\Rightarrow \qquad \alpha^5=3\alpha^2+2\alpha=3(\alpha+1)+2\alpha=5\alpha+3$

Similarly, $\beta^5=5\beta+3$

So $\alpha^5+\beta^5=5(\alpha+\beta)+6=5(1)+6=11$

2.(A) Clearly common ratio $r\in(0,1)$

$x=s_\infty=\dfrac{a}{1-r}=\dfrac{5}{1-r}$

$o<r<1 \Rightarrow o>-r>-1 \Rightarrow 1>1-r>o \quad \Rightarrow 1<\dfrac{1}{1-r}<\infty \quad \Rightarrow 5<\dfrac{5}{1-r}<\infty \Rightarrow 5<x<\infty$

3.(A) $f(x)=x^2+2(p-3)x+9$ Clearly $f(6)<0 \Rightarrow p<-\dfrac{3}{4}$ Greatest integral value of p is -1.

4.(D) $(x-A)(x-12)=-2 \Rightarrow (x-A)(x-12)=(-2)(1)=(2)(-1)=(-1)(2)=(1)(-2)$

Case (1) $\quad x-A=-2 \quad , \quad x-12=1 \Rightarrow A=15$
Case (2) $\quad x-A=2 \quad , \quad x-12=-1 \Rightarrow A=9$
Case (3) $\quad x-A=-1 \quad , \quad x-12=2 \Rightarrow A=15$
Case (4) $\quad x-A=1 \quad , \quad x-12=-2 \Rightarrow A=9$

So, two values of A are possible so sum of possible value of A is $9+15=24$

5.(B) $T_9=A+8D=0$

$\dfrac{T_{29}}{T_{19}}=\dfrac{A+28D}{A+18D}=\dfrac{(A+8D)+20D}{(A+8D)+10D}=\dfrac{0+20D}{0+10D}=\dfrac{2}{1}$

6.(C) $\dfrac{x^2+y^2}{x-y}=\dfrac{(x-y)^2+2xy}{x-y}=(x-y)+\dfrac{2}{x-y}$ Let $x-y=t$ Clearly $t>o \because x>y \quad A\cdot M\cdot \geq G\cdot M\cdot$

$\dfrac{t+\frac{2}{t}}{2}\geq\sqrt{t\cdot\frac{2}{t}} \Rightarrow t+\dfrac{2}{t}\geq 2\sqrt{2}$

$\Rightarrow (x-y)+\dfrac{2}{(x-y)}\geq 2\sqrt{2}$ So $\left(\dfrac{x^2+y^2}{x-y}\right)\min=2\sqrt{2}$

7.(A) $T_n=Pn^2+Qn+R$, $T_1=P+Q+R=240 \quad \to(1)$ $,T_2=4P+2Q+R=336 \quad \to(2)$
$,T_3=9P+3Q+R=440 \quad \to(3)$

Solving (1), (2), (3), We get $P=4, Q=84, R=152$ $\;T_{10}=100P+10Q+R=1392$

8.(D) $\dfrac{-D_1}{4A_1}>\dfrac{-D_2}{4A_2} \Rightarrow -\left(\dfrac{A^2-16}{8}\right)>-\left(\dfrac{4+4B-4}{-4}\right) \Rightarrow A^2+8B<16 \qquad 2x^2+Ax+(2-B)=0$ has discriminant $D=A^2+8B-16 \Rightarrow D=A^2+8B-16<0$ Hence roots are complex

9.(B) Let $x^2=t \Rightarrow At^2+Bt+C=o$ must have both roots positive $\Rightarrow B^2-4AC\geq o, \dfrac{-B}{A}>o, \dfrac{C}{A}>o$

$\Rightarrow A, C$ are of same sign and A,B are root of opposite sign

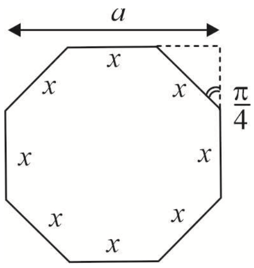
10.(C) Let number are A, B, C, D, E

$2B=A+C,\; C^2=BD \;, C^2=AE$

$2B = A + C \Rightarrow 2\frac{C^2}{D} = \frac{C^2}{E} + C \Rightarrow \frac{2}{D} = \frac{1}{E} + \frac{1}{C} \quad \Rightarrow$ C, D, E are in $H \cdot P \cdot$

11.(C) Each interior angle of a regular octagon $= \frac{(8-2)\pi}{8} = \frac{3\pi}{4}$ Hence $a = x + x\cos\frac{\pi}{4} + x\cos\frac{\pi}{4}$

$a = x + x\sqrt{2} \Rightarrow \frac{x}{a} = \frac{1}{\sqrt{2}+1} = \sqrt{2} - 1$



$S_\infty = \frac{1}{1-\frac{x}{a}} = \frac{1}{1-\left(\sqrt{2}-1\right)} = \frac{1}{2-\sqrt{2}}$

12.(B) α, β are roots of $x^2 - Px + Q = 0,\ \alpha + \beta = P,\ \alpha\beta = Q \quad \alpha, \frac{1}{\beta}$ are roots of

$x^2 - Ax + B = 0,\ \alpha + \frac{1}{\beta} = A, \frac{\alpha}{\beta} = B$

Now $(Q-B)^2 = \left(\alpha\beta - \frac{\alpha}{\beta}\right)^2 = \alpha^2\left(\beta - \frac{1}{\beta}\right)^2 = \alpha\beta \cdot \frac{\alpha}{\beta}\left(\alpha + \beta - \alpha - \frac{1}{\beta}\right)^2 = BQ(P-A)^2$

13.(B) $3a_1 + 7a_2 + 3a_2 - 4a_5 = 0$

$\Rightarrow \quad 7(a_1 + a_2 + a_3) = 4(a_1 + a_3 + a_5)$

$\Rightarrow \quad 7\left(A + AR + AR^2\right) = 4\left(A + AR^2 + AR^4\right)$

$\Rightarrow \quad 7\left(1 + R + R^2\right) = 4\left(1 + R + R^2\right)\left(1 - R + R^2\right)$

$\Rightarrow \quad 7 = 4R^2 - 4R + 4 \Rightarrow R = \frac{3}{2}, \frac{-1}{2}$

14.(C) $2 + 9D = 3 \Rightarrow 0 = \frac{1}{9}$

$2R^9 = 3 \Rightarrow R^9 = \frac{3}{2}$

$A_7G_{19} = \left(2 + \frac{6}{9}\right)2\left(R^9\right)^2 = \frac{8}{3} \times 2 \times \frac{9}{4} = 12$

$A_{19}G_{10} = \left(2 + \frac{18}{9}\right)2\left(R^9\right) = 4 \times 2 \times \frac{3}{2} = 12$

$A_7G_{19} + A_{19}G_{10} = 12 + 12 = 24$

15.(C) $\frac{1}{(1-\alpha)(1-\beta)} + \frac{1}{(1-\beta)(1-\gamma)} + \frac{1}{(1-\gamma)(1-\alpha)} = \frac{1-\gamma+1-\alpha+1-\beta}{(1-\alpha)(1-\beta)(1-\gamma)}$

$= \frac{3-(\alpha+\beta+\gamma)}{1-\alpha\beta\gamma+\alpha\beta+\beta\gamma+\gamma\alpha-\alpha-\beta-\gamma} = \frac{3-\left(-\frac{5}{4}\right)}{1-\left(\frac{-3}{4}\right)+\frac{7}{4}-\left(\frac{-5}{4}\right)} = \frac{17}{19}$

16.(B) $4x^2-8x+3=0$ has roots $\frac{1}{2},\frac{3}{2}$

Case (1) If $x^2+Ax-1=0$ has one root $\frac{1}{2}$ then other root is $-2 \Rightarrow A=\frac{3}{2}$

If $2x^2+x+B=0$ has one root $\frac{3}{2}$ then other root is $-2 \Rightarrow B=-6$

Case (2) If $x^2+Ax-1=0$ has one root $\frac{3}{2}$ then other root is $\frac{-2}{3} \Rightarrow A=\frac{-5}{6}$

If $2x^2+x+B=0$ has one root $\frac{1}{2}$ then other root is -1 $\Rightarrow B=-1$

Clearly case (1) satisfy the require conditions

Hence $12A+B=12\left(\frac{3}{2}\right)+(-6)=12$

17.(C) Case $\to$ 1 $A=1 \Rightarrow s=2008$

Case $\to$ 2 $A\in(0,1), \Rightarrow s=\frac{1-A^{2008}}{1-A}=\frac{1-(2A-1)}{1-A}=2$

Sum of all possible $s=2008+2=2010$

18.(B) Product of roots $=\frac{2A}{3}=1 \Rightarrow A=\frac{3}{2}$

Sum of roots $=e^t+e^{-t}=\frac{A+B}{3}$

A.M. $\geq$ G.M. $\frac{e^t+e^{-t}}{2}>\sqrt{e^t\cdot e^{-t}} \Rightarrow \frac{A+B}{6}>1 \quad (\because t\neq 0)$

$A+B>6= \quad B>6-\frac{3}{2} \Rightarrow B>\frac{9}{2} \Rightarrow$ Least possible B = 5

19.(D) Clearly 9 numbers are of one digit and 90 numbers are of two digits

Let there are x numbers of three digits $9(1)+90(2)+x(3)=2007 \Rightarrow x=606$

Now to find 606th term of 100, 101, 102, … $T_{606}=100+(606-1)(1)=705$

Therefore 2007 the digit is 5

20.(B) $\sqrt{2x^2+3x+5}+\sqrt{2x^2-3x+5}=3x \quad (i)$

$\left(\sqrt{2x^2+3x+5}+\sqrt{2x^2-3x+5}\right)\left(\sqrt{2x^2+3x+5}-\sqrt{2x^2-3x+5}\right)=3x\left(\sqrt{2x^2+3x+5}-\sqrt{2x^2-3x+5}\right)$

$\Rightarrow \quad 6x=3x\left(\sqrt{2x^2+3x+5}-\sqrt{2x^2-3x+5}\right)$

$\Rightarrow \quad \sqrt{2x^2+3x+5}-\sqrt{2x^2-3x+5}=2 \quad \ldots (ii)$

$\because \quad x=0$ is not a solution of given equation from (i), (ii)

$\sqrt{2x^2+3x+5}=\frac{3x+2}{2}$ clearly $\frac{3x+2}{2}$ clearly $\frac{3x+2}{2}$ must be positive

Squaring both sides $4(2x^2+3x+5)=9x^2+4+12x$

$\Rightarrow \quad x^2=16 \Rightarrow x=4,-4$

For $x=-4, \sqrt{2x^2+3x+5}+\sqrt{2x^2-3x+5}=-12$

Not possible so $x=4$ is only one real solution

Integer Type

21.(25) $T_n = \frac{1}{2}\frac{\left(6n^2+2\right)}{\left(n-1\right)^3\left(n-1\right)^3} = \frac{1}{2}\left(\frac{1}{(n-1)^3} - \frac{1}{(n+1)^3}\right)$

$T_2 + T_3 + T_n ... T_n = \frac{1}{2}\left(\frac{1}{1^3} + \frac{1}{2^3} - \frac{1}{n^3} - \frac{1}{(n+1)^3}\right)$

$S = \lim_{n\to 0}\frac{1}{2}\left(1 + \frac{1}{8} - \frac{1}{n^3} - \frac{1}{(n+1)^3}\right) = \frac{1}{2}\left(1 + \frac{1}{8} - 0 - 0\right) = \frac{9}{16} = \frac{p}{q} \Rightarrow p + q = 25$

22.(9) $\frac{x+y}{2} = 2 + \sqrt{xy} \Rightarrow \left(\sqrt{y} - \sqrt{x}\right)^2 = 4 \Rightarrow \sqrt{y} = 2 + \sqrt{x}$

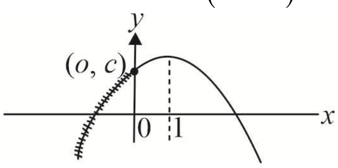
$\left(x, y\right) \Rightarrow \left(1^2, 3^2\right), \left(2^2, 4^2\right), \left(3^2, 5^2\right), \left(4^2, 6^2\right), \left(5^2, 7^2\right), \left(6^2, 8^2\right), \left(7^2, 9^2\right), \left(8^2, 10^2\right), \left(9^2, 11^2\right)$

Total nine pairs are possible.

23.(2) Let $f\left(x\right) = Ax^2 + Bx + C$

$f\left(x\right) - f\left(x+1\right) = Ax^2 + Bx + C - A\left(x+1\right)^2 - B\left(x+1\right) - C = 2x - 1$

$\Rightarrow \quad -2Ax - \left(A+B\right) = 2x - 1 \forall x \in R$



$\Rightarrow \quad -2A = 2, \quad A + B = 1 \quad \Rightarrow \quad A = -1, B = 2$

$f\left(x\right) = -x^2 + 2x + C$

$f\left(x\right)_{\max} = C$ for $x \le o \quad \Rightarrow C = 2$

$f\left(x\right) = -x^2 + 2x + 2$

$f\left(1\right) + f\left(-1\right) = \left(3\right) + \left(-1\right) = 2$

24.(5) Let $C = KB$, $B = \frac{2AC}{A+C} \quad \Rightarrow B = \frac{2KBA}{A+KB} \quad \Rightarrow \quad B = 2A - \frac{A}{K}$

$B = 40 - \frac{20}{K}$

$\because$ B is integer so $k = 2, 4, 5, 10, 20$ are possible

$\Rightarrow$ B = 30, 35, 36, 38, 39

Total five values of B is possible.

25.(5) $D > 0 \Rightarrow (A+B)^2 - (A - B + 8) > 0 \;\; \forall A \in R$

$\Rightarrow \quad A^2 + A(2B-1) + B^2 + B - 8 > 0 \; \forall \; A \in R$

So, discriminant of quadratic in A must be negative

$\Rightarrow \quad (2B-1)^2 - 4(B^2 + B - 8) < 0 \quad \Rightarrow \quad B > \frac{33}{8}$

So least possible natural number B is 5